The Observer–Context Category: Relative Facts, a $\mathbb{Z}_2$ Switching Invariant, and the Movable Cut

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July 2026

Abstract

We make "observer" a mathematical object: an isometric enlargement of the system together with a measurement context on the enlarged algebra, organized into a category (whose invertible switches form a subgroupoid) carrying the $U(1)$ cocycle of the companion papers [2, 3, 4]. Four results follow, each backed by machine certificates and one anchored by superconducting-hardware data. (1) *Relative facts*: every observer admits an exact local classical section, while no global value assignment exists for the Peres–Mermin family — nor for any enlarged family containing its image under a Wigner enlargement. (2) A gauge-robust $\mathbb{Z}_2$ invariant of switch loops: $+1$ on all loops of Weyl switches, -1 on the oriented mutually-unbiased triangle. Nontriviality *requires* complementary switching; the converse fails, and the counterexample is pinned. The invariant counts τ-level switching parity — the same object as the Wigner–friend $-i$. (3) *The movable cut*: the comparison map between cut placements is fixed only up to section; its ray-level content is invariant (and measured), the section-level bookkeeping shifts by the loop holonomy, and the update rule available to each observer is itself forced. (4) The Frauchiger–Renner chain is *localized*: its observer cycle has trivial class and exactly invariant holonomy $+1$ — refuting the natural conjecture that such paradoxes measure holonomy — while its contradiction is state-sector, with contextual fraction exactly $1/6$ and the CHSH inequality at value $7/3$ as the unique separating facet of the codeword polytope. Observer-dependence, the triangle invariant, and cut conventionality are thereby organized by one cohomological object; we present them as structural characterizations within the stated axioms, not an interpretation-free derivation of Copenhagen.

1 Introduction

The Copenhagen principle of the companion note asserts that quantum theory is locally valid — classical within every measurement context — and globally obstructed. This paper extends that structure from contexts to *observers*. The blocking issue in earlier attempts was that "observer" remained informal. We repair this with a minimal definition sufficient for the core theorems formalizing the relativity of facts, the holonomy content of complementary switching, and the movability of the Heisenberg cut. The finite certificates and numerical checks are supplied in the public verification suite (`observer_groupoid.py` and companions), while analytic proofs and interpretive consequences are labeled separately; the holonomy split of Section 4 carries a hardware measurement. Concurrent work benchmarks Wigner's-friend circuits on superconducting hardware from an operational standpoint [5], and the measurement-problem no-go theorems have been sharpened in [6]; the present categorical construction is complementary to both.

2 The observer–context category

Fix the finite Weyl model at dimension d with m registers: observables are Weyl operators $W(v)$, a *context* is a closed commuting Weyl family (not necessarily maximal), and the section $W(u)W(v) = \tau^{c(u,v)}W(u+v)$, $\tau = e^{i\pi/d}$, defines the 2-cocycle c of the companion papers [2, 4].

Definition 1 (Observer; morphisms)**.** *An* observer *is a pair $o = (\iota, C)$: an isometric enlargement $\iota : \mathcal{H} \to \mathcal{H} \otimes \mathcal{H}_O$ (the* cut datum*) and a context C on the enlarged algebra. Morphisms are generated by (i)* switches*: unitaries of a fixed enlargement carrying C to C', on which the Weyl-section gauge group acts by left and right multiplication; and (ii)* refinements*: further isometries $(\iota, C) \to (\iota' \circ \iota, C \otimes I)$ comparing cut placements. Observers and these morphisms generate the* observer–context category $\mathbb{O}$*: the switches are isomorphisms and form a subgroupoid, while the refinement isometries into larger Hilbert spaces are non-invertible, so $\mathbb{O}$ is a category (not a groupoid), and the holonomy invariants below are taken on the switch subgroupoid.*

Lemma 1 (Pullback and monotonicity)**.** *$A \mapsto A \otimes I_O$ is an injective strict morphism of context structures pulling the cocycle back on the nose; in particular operator products of embedded contexts are unchanged, for* any *enlargement. Hence if a family admits no global value assignment, neither does any enlarged family containing its image: an assignment would restrict to one for the embedded copy. Certificate: the six embedded Peres–Mermin context products at $d = 8$ are $(+1)^{\times 5}, (-1)^{\times 1}$, identical to $d = 4$ (`observer_groupoid.py`, part A).*

3 Theorem 1: relative facts

Theorem 1. *(a) Every observer admits an exact local classical section: the cocycle restricted to any single context is a coboundary, with an explicit trivialization λ valued in μ_d for even d and μ_{2d} for odd d (sharp at $d = 3$); verified at $d = 2$ (all six PM contexts), $d = 3$, and $d = 4$ (`local_classicality.py`, `mu_d_valuation.py`). The earlier μ_{4d} / even-d half-τ-step wording is retracted (V46). (b) No global value assignment exists for the PM family, hence none for any enlarged family containing its image (PM product certificate plus Lemma 1).*

Each observer's facts are classical; there is no observer-independent assignment of facts. Observer-dependence is the local-to-global cohomological gap — a theorem, not an interpretive gloss. The trivializations λ are the observer's classical bookkeeping, and their mutual incompatibility around obstructed families is the entire content of contextuality at the observer level.

4 A $\mathbb{Z}_2$ switching invariant

For a loop of switches $U_k \cdots U_1$ closing on its initial context, define the section holonomy $\mathrm{hol} = \det(W_c U_k \cdots U_1)$ with W_c any SU closer. The value hol itself is section-dependent (the lesson of the companion paper's Wigner–friend analysis); its square is not:

Lemma 2 (Invariance — proved)**.** *Every Weyl element $w = i^k P$ has $\det(w)^2 = 1$, and closers are special unitary. Hence Weyl regauging of any switch multiplies* hol *by ± 1, and* hol^2 *is independent of regauging and of the closer. (Determinant fact machine-checked over all* 16 *Weyl elements.)*

Theorem 2. *At $d = 2$: (i) every loop of Weyl switches has* $\mathrm{hol}^2 = +1$*; (ii) the oriented mutually-unbiased triangle $T \to T_x \to T_y \to T$ (switches H, S) has* $\mathrm{hol}^2 = -1$*, with* $\mathrm{hol} = -i$ *in the Clifford presentation. In that presentation* $\mathrm{hol}^2 = (-1)^{\#S \bmod 2}$*: the invariant is the parity of τ-level switches. Certificates: 200 Weyl regaugings and 500 $SU(2)$ closers, single values throughout.*

Corollary 1. $\mathrm{hol}^2 = -1$ *requires complementary switching: loops confined to Weyl switches (in particular, context-preserving ones) are trivial.*

Remark (The converse fails — pinned)**.** *The 2-cycle $T \to T_x \to T$ with switches (H, H) traverses a complementary pair yet has* $\mathrm{hol}^2 = +1$ *(machine-pinned). The invariant therefore detects the* oriented MUB triangle *— the triality that produces the Wigner–friend $-i$ — not pairwise complementarity. We state this openly; "complementarity is holonomy" in the equivalence sense is false, and necessity (Corollary 1) is the correct claim.*

Hardware: on `ibm_fez` the ray-level holonomy of the triangle loop measures $+1$ (loop phase $-2.9^\circ \pm 1.6^\circ$), excluding $-i$ at ray level by $\sim 56\sigma$ (shot-noise statistical, under the implemented interferometric model) — exactly the section split the theorem requires.

Conjecture 1. *In the Clifford presentation at general d,* $\mathrm{hol} \in \mu_{2d}$ *and the Weyl-gauge-invariant residue* $\mathrm{hol}^2 \in \mu_d$ *detects the top layer of the 2-adic tower of the companion papers.*

5 Theorem 3: the movable cut

Theorem 3. *Let two observers differ only by a refinement (friend versus Wigner descriptions of one run). The comparison map between their descriptions is fixed only up to the section choice: its ray-level content is invariant — and is what the hardware measures — while the section-level bookkeeping shifts by the loop holonomy of Section 4. Three levels must be kept apart: the empirical probabilities are invariant under the move; the determinant-*section *phase shifts by the loop holonomy; and the cohomology class is unchanged. In this precise sense ray-level physics is independent of the cut placement while the section bookkeeping records the move; the broader reading "the cut may be moved without empirical consequence" is the interpretation this supports, not a claim beyond the three-level statement. Moreover the update rule available to each observer is constrained (not a free convention): statistics, repeatability, and commutant covariance force the one-parameter affine depolarizing family per degenerate block (complete-positivity interval $-1/(r^2 - 1) \le p \le 1$), with the Lüders instrument the unique single-Kraus member (`lueders_instrument.py`, a numerical stress test; CP interval certified in `lueders_cp_interval.py`).*

This is consistent with Copenhagen's conventionality of the cut, now given structural content: the cut is a choice of trivialization, ray-level physics is invariant under the choice, and the invariant that survives is precisely the class. We offer this as a structural reading, not a claim that it is the only interpretation the formalism admits.

6 Theorem 4: the Frauchiger–Renner cycle, localized

Model each lab by one qubit (memory identified with the measured system, as is standard); the probabilistic core of the FR chain is Hardy's structure on $\psi = (|h, \downarrow\rangle + |t, \downarrow\rangle + |t, \uparrow\rangle)/\sqrt{3}$

with the four observer contexts $\{\bar{Z}Z\}, \{\bar{Z}X\}, \{\bar{X}X\}, \{\bar{X}Z\}$ — a closed 4-cycle Weyl family in $\mathbb{O}$ (`fr_cycle.py`).

Theorem 4. *(i) The FR observer cycle has trivial class (all four context phases* $+1$*; the value system is solvable,* $|L| = 16$*) and its switch loop closes exactly with* $\mathrm{hol} = +1$*; since every two-qubit Weyl element has determinant* $+1$*, this holonomy is fully gauge-invariant. The FR paradox is therefore* not *a holonomy phenomenon: the natural conjecture that observer-cycle paradoxes measure holonomy is refuted by the machinery of this program itself. (ii) The paradox is state-sector:* $\mathrm{CF}(F_{\mathrm{FR}}, \psi) = 1/6$ *exactly, and by the codeword-polytope theorem of the companion paper [4] the FR contradiction is equivalent to the moment vector of* ψ *lying outside* $\operatorname{conv} L$*. The separating facet, extracted by exact separation, is the CHSH inequality* $\langle \bar{Z}Z\rangle - \langle \bar{Z}X\rangle + \langle \bar{X}X\rangle + \langle \bar{X}Z\rangle \le 2$ *with quantum value* $7/3$*, and the Abramsky–Barbosa–Mansfield normalization reproduces the LP value:* $\mathrm{CF} = (7/3 - 2)/2 = 1/6$*.*

The Hardy reading of FR is known; the contribution here is its exact placement in the program's taxonomy. Consistency conditions for observer networks are polytope-membership conditions in the state sector, not holonomy conditions: "absoluteness of observed events" fails where the codeword polytope is exited, at a quantified rate.

7 Discussion and open problems

Open: the general-d tower refinement of Section 4; multi-round dynamics beyond a single update; whether any observer-network paradox is genuinely holonomy-sector (Theorem 4 says FR is not — a candidate would need an odd cycle, i.e. an AvN core); and external review beyond the author's own adversarial checks.

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